

Sustainability or Greenwashing: Evidence from the Asset Market for
Industrial Pollution
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October 6, 2025

Research Question

- the reallocation of industrial pollution in response to environmental pressures through acquisitions and sales of pollutive assets in the real asset market.
 - how firms respond to environmental pressures
 - which firms buy and sell pollutive assets
 - how pollution levels change around the transfer of ownership
 - what gains firms realize from trading these assets
- broader: how effective divestment is

In this paper (1)

- a conceptual framework that considers heterogeneous firms holding pollutive assets and facing varying levels of environmental pressures
 - firms can have various responses depending on
 - (1) the intensity of the environmental pressures they face and,
 - (2) on the relative costs of benefits of operating pollutive assets versus the alternatives
- construct measures of the env. pressures that firms face, including
 - a firm's coverage by ESG rating agencies,
 - pension fund holdings,
 - the political climate where the company is headquartered,
 - the frequency of environmental risk incidents, and
 - a composite index that is the average of the preceding measures.

estimate choice models that evaluate firms' responses to environmental pressures

- divestment is a **prominent** response to environmental pressures
 - 1 SD increase in the composite environmental pressure index → 54% increase in divestment likelihood (e.g. plant closure or abatement efforts: 9% to 31%)

In this paper (2)

- focus on divestitures of pollutive industrial plants from 2000 to 2020: investigate both their determinants and implications for buyers and sellers
 - study the decision to sell and buy pollutive assets
 - firms are more likely to divest pollutive assets under stronger environmental pressures, and these effects are considerably more pronounced when pollution levels are higher.
 - compared to the sellers, buyers of pollutive assets face significantly weaker environmental pressures
 - a separating real-asset-market equilibrium whereby public firms with mounting environmental pressures sell their most pollutive assets to firms with weaker pressures
- examine changes in pollution around divestitures
 - no difference between the change in pollution at divested and not divested plants
 - no difference between pollution abatement efforts at sold versus unsold plants
 - nonrandom divestiture? trace the combined pollution levels of sellers' and buyers' plants around divestitures
 - no reduction in the pollution levels of sellers' and buyers' combined plants
 - obsolescence or capital reallocation? → no difference in productivity growth and survival rates; no introduction of new plants

In this paper (3)

- selection and omitted variables: focus on divestitures that follow quasi-exogenous environmental risk incidents → no changes in pollution following those divestitures and no pretrends in pollution before they occur

⇔ **“greenwashing” divestment strategy to mitigate stakeholder pressures**

- role of information costs as an indirect proxy for the potential for deception or misalignment between corporate managers and outside stakeholders
→ firms with complex organizational structures and more dispersed ownership are more likely to respond to environmental pressures by divesting pollutive assets
- divested assets are sold to firms that have preexisting business ties or develop new business ties with the sellers following the sale; more likely when the connected firms face weaker environmental pressures (greenwashing: sellers maintain their access to the sold plants)

In this paper (4)

- Bidirectional Encoder Representations from Transformers language model to analyze the text of firms' conference calls with investors
 - following divestitures of pollutive plants, sellers are considerably more likely to mention and emphasize improvements in their environmental policies (advertisement)
- investigate the gains from trading pollutive assets: following pollutive asset divestitures,
 - sellers' ESG ratings increase by roughly 103% relative to the sample average, with a particularly strong improvement for environmental ratings (160% relative to the sample average)
 - the likelihood of an EPA enforcement action drops
 - the costs of regulatory enforcement, including fines and cleanup costs, also decline considerably

present only following the divestment of pollutive assets, not a general feature of asset sales

In this paper (4)

- Do shareholders recognize the above benefits from offloading pollutive assets?
 - estimate sellers' cumulative abnormal returns (CARs) around the announcement of pollutive asset divestitures
 - average CAR is significantly higher when the divested plant is more pollutive and when the divestitures are accompanied by positive environmental disclosures
- market-based evidence that the buyers of pollutive assets gain from these trades by paying discounted prices
 - the gains of the buyers relative to the sellers increase with pollution levels of the transferred assets.

Data and Variables (1)

to measure the environmental pressures of a firm:

- public listing status (subset)
- ESG rating coverage — KLD
- ESG rating changes over time: CSR score = netting the total number of strengths and weaknesses across all categories; separately Env Score — KLD
- pension holdings (1 if above average) — 13F
- local political ideologies: county-level vote share data → Democratic HQ (whether a firm's headquarters is located in a Democratic-leaning county) — MIT Election Data and Science Lab
- occurrence of negative environmental incidents: Env. Event (1 if incurring an env event) — Reprisk
- Composite index: Pressure Index = average across all indicators that are available for a firm-year

Data and Variables (2)

Pollution and Abatement:

- chemical-level toxic emissions for each plant from the EPA's TRI program over the period 2000 to 2020 (mandatory reporting for establishments satisfying some criteria): measures of toxic release at the plant-chemical level, total & intensity
- EPA's pollution abatement activities: source reduction activities — #Source Reduction, postproduction waste management activities — % of total generated toxic waste reduced through: % recycling, % recovery, % treatment
- accounting: CRSP, string-matching algorithm based on ultimate parent company name

Divestitures:

- SDC completed M&A between 2000 and 2020: only divestiture/spin-off + buyer owning > 50% after the deal, non-financial (nor financial subsidiary) buyer and seller matched to divested TRI plants $\implies$ 888 deals involving 1105 unique plants & divestitures of nonpollutive assets over 2000-2020 (those not linked to TRI)

Data and Variables (3)

Enforcement Actions and Compliance Costs:

- Enforcement and Compliance History Online: costs (in dollars) of federal and local penalties, compliance, recovery, and supplemental environmental projects.
→ total regulatory compliance costs

Conference Call Transcripts:

- Thomson Reuters's SE database starting in 2001: management presentation, use a machine learning algorithm to identify the language related to environmental disclosure and the associated tone → Positive Env. Disclosure & Negative Env. Disclosure

Supply Chain and Joint Venture Relationships:

- measure business ties: Factset and Compustat Segment databases + SDC → 1 if the acquirer and the target share either a supply chain or a joint venture connection

CAR:

- CSRP; three-day window, three-factor model
- division of surplus between the buyer and the seller: the difference between the change in the buyer's market value of equity and the change in the seller's market value of equity in the three-day window around the deal's announcement (*product*)

Sellers and Buyers of Pollutive Assets

investigate the motives behind the divestment of pollutive plants, and characterize the buyers and sellers of pollutive plants:

- multilogit regression framework in which firms can respond to environmental pressures:
 - divestment of pollutive plants,
 - closure of pollutive plants without divestment or enhanced pollution abatement,
 - enhanced pollution abatement without closure or divestment
 - both plant closures and enhanced pollution abatement without divestment.

The omitted benchmark is no action.

→ Table II

Sellers of Pollutive Assets

- regressions that explain the propensity to divest pollutive plants: Eq.(1), p18 (env. pressures) → Table III A
- correlation with operational inefficiencies? the role of pollution levels in divestment amid environmental pressures → Table III B
- the likelihood of divesting pollutive plants (i.e., Sell (Pollutive)) in which the key independent variable is the occurrence of environmental risk incidents. → Table IV
- the timing of divestitures surrounding the occurrence of environmental risk incidents and the possibility of preevent trends;
a dynamic model that tracks how a negative environmental incident influences the likelihood of divestment each year before and after the incident. → Figure 2

Who are the buyers of these assets, and whether they have a comparative advantage in operating and owning pollutive assets?

- compare buyers' and sellers' exposure to environmental pressures. → Table V

Changes in Pollution around Divestitures

- estimate generalized DID regression specifications using two-way fixed effects, Eq(2) p24
- address concerns related to heterogeneous treatment timing effects in generalized DID regressions by estimating stacked event regressions
match each sold plant to never-sold plants in the same industry → a “cohort”.
stack all such cohorts → make within-cohort comparisons

Table VI

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Conclusion

Reference I